

Homework problems:

Exercise 10.1 (10 points). Answer all the questions from the Quiz **Linear Regression**.

*The quiz is available from June 15, 19:00 until **June 22, 09:00**. You have 60 minutes to answer the questions.*

Remark: The points for this problem count for the grade bonus. **Be aware, that you have to use R during the quiz.**

Solution: To see the solution to this problem, please reopen the quiz and click on *Review*.

Exercise 10.2. Suppose we gather data for a group of students in a statistics class with variables X_1 for hours studied, X_2 for undergrad GPA, and Y for receiving an A.

We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

- Compute the probability, using the fitted model, that a student who studies for 40 hours and has an undergrad GPA of 3.5 will get an A in the class.
- How many study hours are needed for a student with an undergrad GPA of 3.5 to achieve a 0.5 probability of getting an A in the class based on the fitted model?

Solution:

- The fitted logistic model is given by

$$\log\left(\frac{p(\mathbf{X})}{1-p(\mathbf{X})}\right) = -6 + 0.05 \cdot X_1 + X_2.$$

or

$$p(\mathbf{X}) = \frac{e^{-6+0.05 \cdot X_1 + X_2}}{1 + e^{-6+0.05 \cdot X_1 + X_2}}.$$

From the second equation we get for $x_1 = 40$ and $x_2 = 3.5$ a probability of $p(X) = 0.38$.

- We would like to solve the equation

$$\log\left(\frac{p(\mathbf{x})}{1-p(\mathbf{x})}\right) = -6 + 0.05 \cdot x_1 + x_2$$

for x_1 with $x_2 = 3.5$ and $p(\mathbf{x}) = 0.5$. Inserting the values for x_2 and $p(\mathbf{x})$ we get

$$\begin{aligned} \log\left(\frac{0.5}{0.5}\right) &= \log(1) = 0 = -6 + 0.05 \cdot x_1 + 3.5 \\ \Leftrightarrow 2.5 &= 0.05 \cdot x_1 \quad \Leftrightarrow \quad x_1 = 50. \end{aligned}$$

Exercise 10.3. Examine the `penguins` dataset. The objective is to apply logistic regression to investigate the relationship between species, flipper length, and the classification of a penguin as “large.” Define a “large” penguin as one with a body mass exceeding the sample average. Begin by removing missing values and creating a new response variable that indicates whether a penguin is large.

```
library(tidyverse)
penguins <- penguins |>
  drop_na() |>
  mutate(large = factor(if_else(body_mass > mean(body_mass, na.rm = TRUE),
                                1, 0)))
```

- Visualize the relationship between `large` and `species`. Summarize the relationship observed between these two variables based on the plot.
- Assume an Adelie penguin is randomly selected from the Adelie penguins contained in the sample. Calculate the probability and the odds that the randomly selected penguin is large.
- Split the data into a 75% training set, `penguins_train`, and a 25% test set, `penguins_test`. Use a seed of 12345. Fit a model using `species` to predict the odds that a penguin is large. Let π denote the probability that a penguin is large. The model is specified as follows.

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{ Chinstrap} + \beta_2 \text{ Gentoo}$$

Fit the specified model using the training data. Determine the predicted odds of a penguin being large for both Gentoo and Adelie species.

- Add flipper length as a predictor to the model. Begin by visualizing the relationship between `flipper_len` and `large`. Summarize the relationship observed between these two variables based on the plot.
- Add the flipper length to the model fitted in c), so that there are two predictors, `species` and `flipper_len`. Based on the fitted model, how do the odds of being a large penguin change when going from a Gentoo penguin with a flipper length of 185 mm to a Gentoo penguin with a flipper length of 200 mm?
- Now, let's evaluate the performance of both models on the test data. Compute the predicted probabilities for being a large penguin using the test data for both models. Use the computed predictions and the function `yardstick::roc_curve()` to compute the ROC curve for both sets of predictions. Plot the ROC curves and comment on the resulting plot.
- Use the fitted ROC curves from the previous part to compute the area under the curve (auc) for both curves. Determine the best threshold (in terms of Youden's J statistic) and the corresponding values for the specificity, sensitivity, and accuracy. Comment on the results you obtain.
Hint: The computed ROC curves contain, in addition to the specificity and sensitivity values, also the corresponding thresholds.

Remark: If you want to, you can use the Quarto template `starter-large_penguins.qmd`, which is available on Moodle, to solve this exercise.

Solution:

We start by loading the tidyverse.

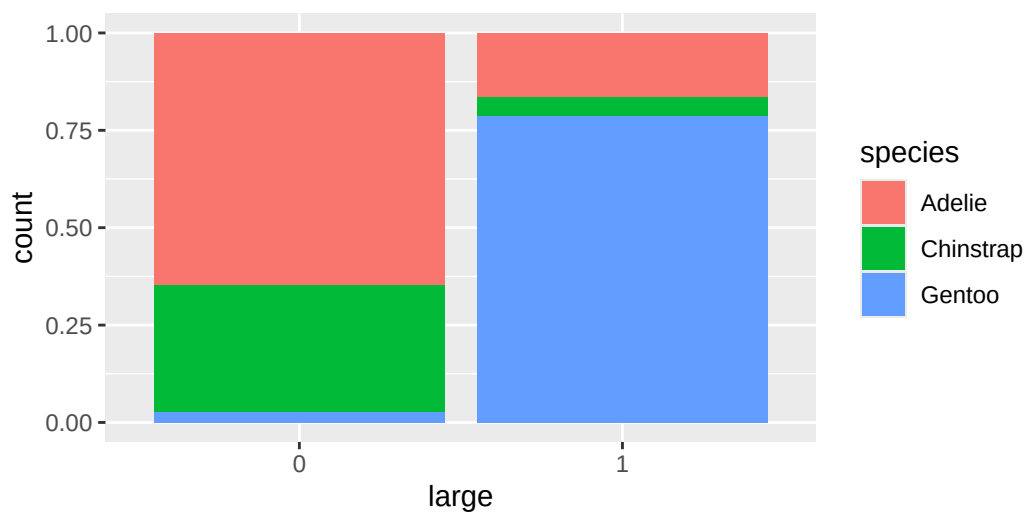
```
library(tidyverse)
```

Now we can do the required data preprocessing, which includes removing all observations with at least one missing value and creating the response variable **large**.

```
penguins <- penguins |>
  drop_na() |>
  mutate(large = factor(if_else(body_mass > mean(body_mass, na.rm = TRUE), 1, 0)))
```

- a) Let's start by examining the relationship between **large** and **species**. Both variables are categorical. So, we can make a filled barplot for **large** and fill the bars with respect to **species**.

```
ggplot(penguins, aes(x = large)) +
  geom_bar(aes(fill = species), position = "fill")
```



The majority of large penguins are Gentoo, resulting in significant differences in the proportions of various species among small and large penguins. So, there is a strong relationship between the two variables.

- b) In this step we consider only the Adelie observations. Hence, we filter out all Chinstrap and Gentoo observations. After doing so, we can compute the absolute and based on that, the relative frequencies for small and large penguins.

```
penguins |>
  filter(species == "Adelie") |>
  count(large) |>
  mutate(p = n / sum(n))
```

	large	n	p
1	0	122	0.8356164
2	1	24	0.1643836

Hence, the probability of selecting a large Adelie penguin from the Adelie penguins contained in the sample is around 0.164.

The odds of being large for an Adelie penguin is then roughly

$$\frac{0.164}{1 - 0.164} = 0.1961722.$$

The exact value can be determined using the following code.

```
penguins |>
  filter(species == "Adelie") |>
  count(large) |>
  mutate(p = n / sum(n)) |>
  summarize(odds = p[2] / p[1])
```

```
      odds
1 0.1967213
```

- c) We split the data into a training and test part using functions from `tidymodels`. Since this is a random split, we use a seed, to make the results reproducible. The split is computed using `initial_split()` from the `rsample` package. Therefore, we load in a first step `tidymodels` (since we will need further functions from other packages contained in `tidymodels` later on).

```
library(tidymodels)
set.seed(12345)
penguins_split <- initial_split(penguins, prop = 0.75)
penguins_train <- training(penguins_split)
penguins_test <- testing(penguins_split)
```

The given model can be fitted to the training data by using the following code.

```
model_s <- glm(large ~ species, data = penguins_train, family = binomial)
```

We have used the `glm()` function, since we want to fit a logistic regression model, which is specified by choosing `family = binomial`.

The estimated coefficients are the following.

```
tidy(model_s)
```

```
# A tibble: 3 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)   -1.72      0.263    -6.53 6.41e-11
2 speciesChinstrap -0.419    0.506    -0.830 4.07e- 1
3 speciesGentoo    4.43      0.532     8.33 8.10e-17
```

We can use those estimates to calculate the fitted probabilities for a Gentoo or Adelie penguin to be large, which are necessary to determine the predicted odds. But we can also simplify this step by using the `predict()` function.

```
pi_g <- predict(model_s, newdata = data.frame(species = "Gentoo"),
  type = "response")

pi_a <- predict(model_s, newdata = data.frame(species = "Adelie"),
  type = "response")
```

The predicted odds are then give by:

```
pi_g / (1-pi_g)
```

```
1  
15
```

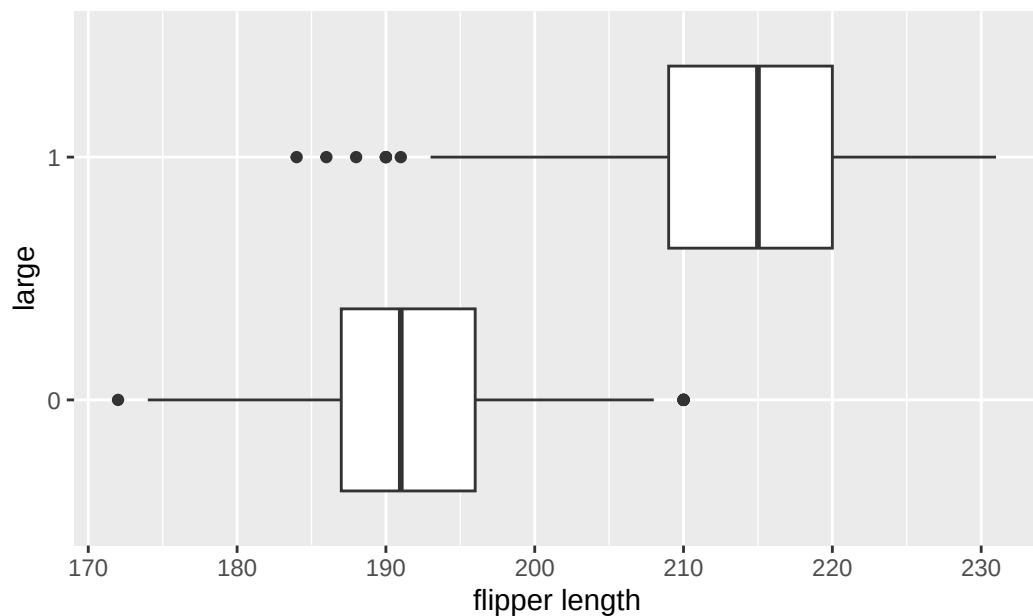
```
pi_a / (1-pi_a)
```

```
1  
0.1789474
```

Hence, the chance of a Gentoo penguin being large is huge, as indicated by the bar plot in a). For an Adelie penguin the story is different.

- d) The variable `flipper_len` is numeric, so a boxplot is suitable to visualize its relationship with the `large` variable across its two levels.

```
ggplot(penguins, aes(y = large)) +  
  geom_boxplot(aes(x = flipper_len)) +  
  labs(x = "flipper length")
```



The average flipper length of large penguins is about 215 mm, which significantly exceeds that of small penguins. Thus, also the flipper length should be informative for predicting the probability of being a large penguin.

- e) We can add the flipper length to the above model by simply adding `flipper_len` to the right hand side of our formula.

```
model_sf <- glm(large ~ species + flipper_len, data = penguins_train,  
                family = binomial)
```

The estimated coefficients are

```
tidy(model_sf)
```

```
# A tibble: 4 x 5
```

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1	(Intercept)	-40.3	8.02	-5.02	0.000000514
2	speciesChinstrap	-1.83	0.661	-2.76	0.00574
3	speciesGentoo	0.140	0.913	0.153	0.878
4	flipper_len	0.200	0.0413	4.85	0.00000124

The relative change in the odds of being large when increasing the flipper length from 185 to 200 mm and keeping `species` equal to Gentoo, is equal to the odds ratio

$$e^{\hat{\beta}_3} \approx e^{0.2 \cdot 15} = 20.0855369.$$

But the flipper length of 185 mm is very atypical for a Gentoo penguin. Hence, the odds for being a large Gentoo penguin with a short flipper length of 185 mm is small.

```
coef_sf <- coef(model_sf)
pi_g <- predict(model_sf, newdata = data.frame(species = "Gentoo",
                                              flipper_len = 185),
               type = "response")

pi_g / (1 - pi_g)
```

```
1
0.045448
```

But also 200 mm is less than the minimal flipper length observed for a Gentoo penguin.

```
penguins |>
  filter(species == "Gentoo") |>
  select(flipper_len) |>
  pull() |>
  summary()
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
203.0	212.0	216.0	217.2	221.5	231.0

Therefore, it makes sense that the chances of being a large penguin due to an increase of 15 mm in flipper length are still small.

```
exp(coef_sf[4]*15) * (pi_g / (1-pi_g))
```

```
flipper_len
0.9169224
```

- f) We already fitted both models to the test data. Hence, we can continue by computing the predictions for the test data.

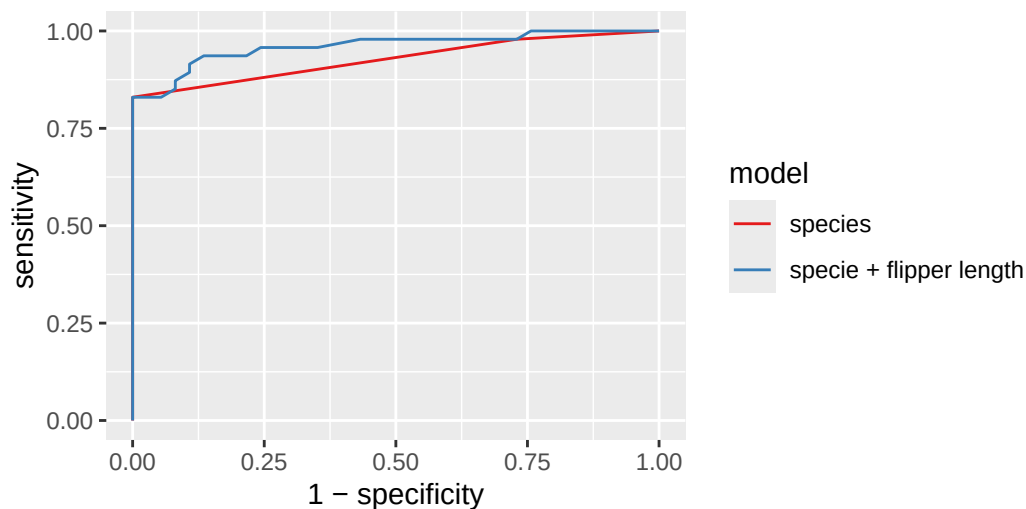
```
penguins_test$.pred_s <- predict(model_s, newdata = penguins_test,
                                type = "response")
penguins_test$.pred_sf <- predict(model_sf, newdata = penguins_test,
                                type = "response")
```

Given those predictions, we can compare them with the actually observed values of `large` in the test data. Based on that comparison we can compute the ROC curve using `roc_curve()` from the `yardstick` package.

```
roc_s <- roc_curve(penguins_test, large, .pred_s,
                  event_level = "second")
roc_sf <- roc_curve(penguins_test, large, .pred_sf,
                  event_level = "second")
```

The two ROC curves are then plotted into one figure with the `ggroc()` function

```
bind_rows(roc_s, roc_sf, .id = "model") |>
  group_by(model) |>
  ggplot(aes(x = 1 - specificity, y = sensitivity, color = model)) +
  geom_path() +
  scale_color_brewer(palette = "Set1", name = "model",
                    labels = c("species", "specie + flipper length"))
```



The ROC curve for the model using both variables is almost always above or the one using only `species`. This indicates that the second model often has a sensitivity as good as `model_s` (or better) for a given specificity.

- g) In f) we already argued that the shape of the ROC curves indicates that we should favour `model_sf`. We can also summarize the shapes by computing the area under the curves.

```
roc_auc(penguins_test, large, .pred_s,
        event_level = "second")
```

```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>    <chr>        <dbl>
1 roc_auc binary      0.927
```

```
roc_auc(penguins_test, large, .pred_sf,
        event_level = "second")
```

```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>       <dbl>
1 roc_auc binary      0.961
```

The larger area under `roc_sf` confirms our finding. So, we would prefer the model using both variables. But since the two ROC curves intersect, the question remains, if we could still achieve the same predictive accuracy with smaller model. This means, can we select thresholds for both model to classify the predicted probabilities such we get the same Youden's J.

To do so, we compute Youden's J statistic from the fitted ROC curves.

```
roc_s |>
  # Calculate Youden's J for every row in the ROC curve
  mutate(j_index = sensitivity + specificity - 1) |>
  # Grab the row where J is maximized
  slice_max(j_index, n = 1, with_ties = FALSE)
```

```
# A tibble: 1 x 4
  .threshold specificity sensitivity j_index
      <dbl>         <dbl>         <dbl>   <dbl>
1      0.937           1         0.830   0.830
```

```
roc_sf |>
  # Calculate Youden's J for every row in the ROC curve
  mutate(j_index = sensitivity + specificity - 1) |>
  # Grab the row where J is maximized
  slice_max(j_index, n = 1, with_ties = FALSE)
```

```
# A tibble: 1 x 4
  .threshold specificity sensitivity j_index
      <dbl>         <dbl>         <dbl>   <dbl>
1      0.820           1         0.830   0.830
```

For the best choice we see that both models yield the same sensitivity and specificity values.

Exercise 10.4 (5 points).

Grade all three submissions assigned to you (one being your own) in the peer-review workshop for Problem H.9.1 by June 22, 09:00.

You find the submissions by re-opening the peer-review workshop Problem H.9.1.